The salinity of the Ocean

Almost every known chemical element can be found in varying proportions in the ocean.

*Salinity is the amount of salt dissolved in a body of water. On average, seawater in the world's oceans has a salinity of approximately 35‰ or 35 parts per thousand. This means that for every 1 kg (1000 grams) of seawater there are 35g of salt dissolved in it.

*Lands are the main source of salinity in the oceans and rivers are the main carriers. Other factors responsible for ocean salinity are - evaporation, wind, river water, precipitation, ocean currents and volcanoes etc.

*During an exploration of **Challenger Deep** in 1884, Dittmar discooured that there are 47 types of salts present in the seas. Out of these 47 types of salts, 7 are the important ones. They are as follows.

Salt	Salinity (‰)	Percentage (%)
Sodium Chloride	27.213	77.8
Magnesium Chloride	3.807	10.9
Magnesium Sulphate	1.658	4.7
Calcium Sulphate	1.260	3.6
Potassium Sulphate	0.863	2.5
Calcium Carbonate	0.123	0.3
Magnesium Bromide	0.076	0.2

*The maximum contribution to the salinity of the ocean is that of sodium chloride.

*The salinity of the ocean can be divided into horizontal and vertical parameters. The uncertainty regarding the amount of salinity in the vertical dimension has obstructed the devlopment of a principle for measuring vertical salinity. This is because in some places, salinity increases with depth and vice-versa.

*The density of ocean water is directly proportional to its salinity. This means that if the density of water increases it is an indicator of an increase in salinity hence with the increase in water density, salinity also increases.

*The horizontal distribution of salinity in ocean water is expressed as the degree of saltiness of the water, either as a percentage or more often in parts per thousand Variations are shown in salinity distribution maps by isolines (lines joining places having an equal degree of salinity). Latitudes have a huge impact on the salinity of the oceans. Salinity is lower than the average **34‰** in equatorial waters because of the **heavy daily rainfall** and **high evaporation**. Deserts between 20º and 40º N and S, waters have high salinity because of the high rate of evaporation caused by high temperatures and low humidity. Waters of the polar region have a salinity of less than 35‰ due to the colder climate with little evaporation and because much fresh water is added from the melting of icebergs as well as by several large poleward-bound rivers, e.g. Ob, Lena, Yenisei etc.

*Maximum salinity is found between 20º-40º North Latitudes where salinity may increase up to 36‰.

*Ocean water of the Indian Ocean falling between 0º-10º Latitudes has a salinity of 35.14‰ however it decreases to 30‰ in Bay of Bengal. On the contrary the salinity increases upto 36‰ in the Arabian Sea. This is due to a high rate of evaporation and low humidity. Additionally there is much less fresh water carried to the Arabian Sea by west flowing rivers as compared to Bay of Bengal.

*Among the various seas and Lakes of the world, the **Van Lake** in Turkiye has a salinity of 330‰. The salinity of Dead Sea is 238‰.

*Salinity in the Northern part of **Caspian Sea** is 13-14‰, whereas in the southern part of Caspian sea, in the Karabuga Bay, salinity is 300-350‰.

*The most saline sea is Red Sea with a salinity of 36-41‰.

*The salinity of the **Mediterranean Sea** is 37-39‰. In the Caribbean Sea it is 35-36‰ and the salinity of the Persian Gulf is 37-40‰.

Note - World's lakes with maximum salinity **Lake Assal** (348‰), Lake Van of Turkiye (330‰), Dead Sea (238‰) and **Great Salt Lake** (220‰).

*The world's most hyper-saline lake is Don Juan Pond in Antarctica, and second to it is Garabogazkol (Turkmenistan), which is a lagoon. It is important to mention here that lake Van of Turkiye is comparatively less saline however the Lake water is strongly alkaline (pH 9.7-9.8).

1. Which one of the following denotes water salinity gradient?

(a) Thermocline
(b) Halocline
(c) Pycnocline
(d) Chemocline

U.P.P.C.S. (Pre) (Re-Exam) 2015

Ans. (b)

Halocline is a zone in oceanic water in which salinity changes rapidly with depth. So, Halocline denotes water salinity gradient.

2. **Main Source of Salinity of the Sea is –**
(a) Rivers
(b) Land
(c) Wind
(d) Ash ejected from the Volcano

53rd to 55th B.P.S.C. (Pre) 2011

Ans. (b)

Main source of the salinity of seawater is the land. While rivers bring various types of salt content to the sea, other factors that regulate the salinity of the sea include vaporization, wind, river water, rain, oceanic currents and volcanoes etc. Average salinity of the sea is about 35‰.

3. **Which one of the following salts contributes maximum to the salinity of sea water :**
(a) Calcium sulphate
(b) Magnesium chloride
(c) Magnesium sulphate
(d) Sodium chloride

U.P.P.C.S. (Pre) 2000

Ans. (d)

In 1884, during Challenger expedition, William Dittmar produced a report about the chemistry of sea water. He revealed in the report that there are 47 types of salts present in the sea water among which 7 are most prominent. These are as follows –

Salt	**Salinity (‰)**	**Percentage (%)**
Sodium Chloride	27.213	77.8
Magnesium Chloride	3.807	10.9
Magnesium Sulphate	1.658	4.7
Calcium Sulphate	1.260	3.6
Potassium Sulphate	0.863	2.5
Calcium Carbonate	0.123	0.3
Magnesium Bromide	0.076	0.2
	35	100

Notably percentage of average salinity in oceans is 35‰ and sodium chloride, plays a major role in the salinity of the sea.

4. **The highest salinity is found in :**
(a) Dead sea
(b) Red sea
(c) Great Salt Lake in the U.S.A.
(d) Lake Van in Turkey

U.P.P.C.S. (Pre) 1995

Ans. (d)

Among the lakes and seas given in the question, Lake Van of Turkiye has maximum salinity which is 330‰.
Salinity in the Dead Sea = 238‰
Salinity in the Red Sea = 36-41‰
Salinity in Great Salt Lake = 220‰

5. **The highest amount of salinity is found in –**
(a) Pacific Ocean (b) Indian Ocean
(c) Mediterranean Sea (d) Dead Sea

M.P. P.C.S. (Pre) 2015
Uttarakhand P.C.S. (Mains) 2006

Ans. (d)

See the explanation of the above question.

6. **Water of which one of the following seas is most saline?**
(a) Baltic Sea (b) Black Sea
(c) Dead Sea (d) Red Sea

Uttarakhand P.C.S. (Pre) 2002
U.P.P.C.S. (Pre) 1998

Ans. (c)

See the explanation of the above question.

7. **World's most saline Ocean/Lake is –**
(a) Caspian Sea (b) Great Salt Lake
(c) Dead Sea (d) Lake Van

R.A.S./R.T.S. (Pre) 1997

Ans. (d)

See the explanation of the above question.

8. **Which of the following Sea has the highest salinity ?**
(a) Caspian Sea (b) Mediterranean Sea
(c) Red Sea (d) Dead Sea

U.P.P.C.S.(Pre) 2012

Ans. (*)

Red Sea has the highest salinity (36 to 41‰). In the above options, the Mediterranean Sea is also a sea with a salinity of 37 to 39‰. The salinity of the Dead Sea is more than that of the Red Sea and the Mediterranean Sea but it is considered as a lake. Caspian Sea is also a lake. If all the options in the question are considered as seas, the Dead Sea has the highest salinity and would be saltiest among these.

9. **When the density in the sea increases, then –**
(a) Salinity and depth decreases.
(b) Salinity increases but depth decreases.
(c) Both salinity and depth increases.
(d) Salinity decreases and depth increases.

U.P. P.C.S. (Pre) 1990

Ans. (*)